温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

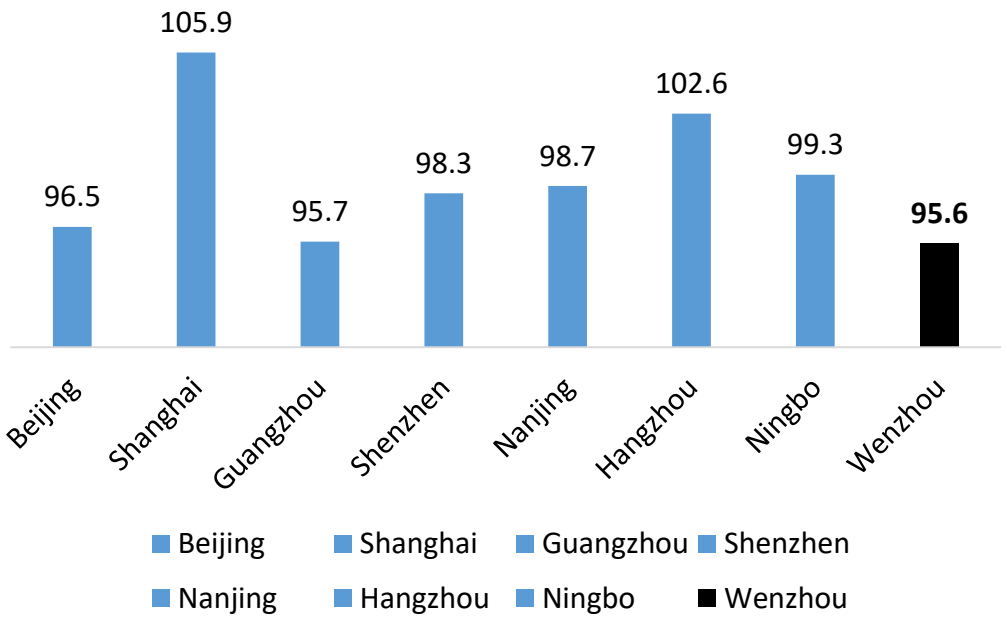
Data Visualization

Dawei Chen

Week 5-1, 30 Sep 2025

Numbers vs. Visuals – Which Tells the Story Better?

City	2025年8月新建商品住宅销售价格指数(上年同月=100)
Beijing	96.5
Shanghai	105.9
Guangzhou	95.7
Shenzhen	98.3
Nanjing	98.7
Hangzhou	102.6
Ningbo	99.3
Wenzhou	95.6



Source: 国家统计局 - 70个大中城市住宅销售价格指数

Which one helps you understand Wenzhou's housing prices faster:

A. The table of numbers
B. The chart/visualization

Outline

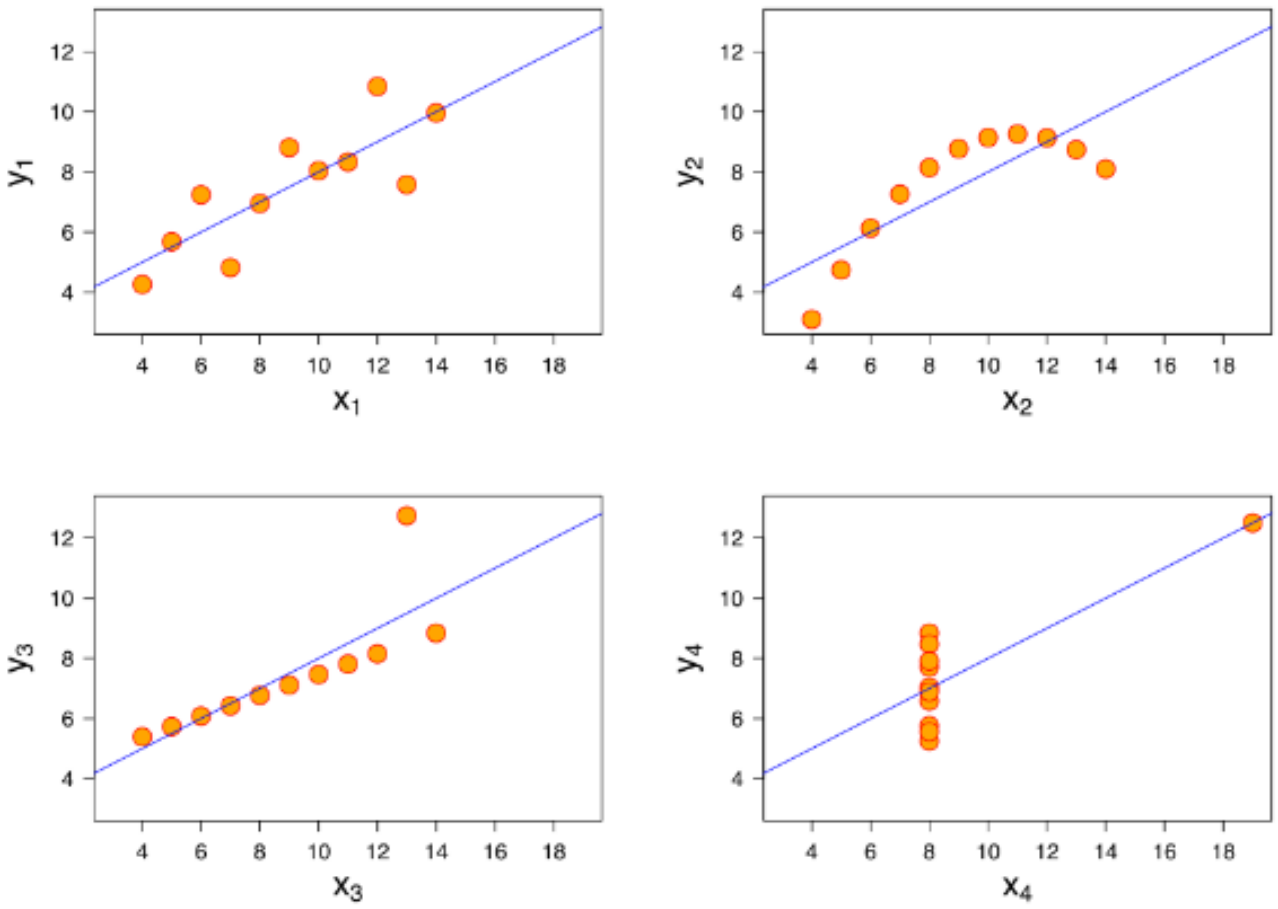
- What & Why Data Visualization
  - Anscombe's Quartet, Datasaurus dozen
- What Makes a Good Data Visualization
  - Principles and examples
- How to Visualize Data
  - Using Design Wisely: preattentive Attributes, data-ink
  - Tables
  - Charts

What & Why Data Visualization?

- Definition
  - Graphical representation of information and data
  - Uses visual elements: **tables**, **charts**, **maps**, **infographics**, **dashboards**
- Why It Matters?
  - *Simplifies complexity* → makes large datasets understandable at a glance
  - *Reveals patterns & trends* → see relationships, anomalies, outliers
  - *Improves communication* → share insights with managers, clients, public, **non-experts**
  - *Supports decisions* → visual evidence helps justify actions
  - *Engages audience* → visuals are more memorable than raw numbers

Takeaway: *Data visualization transforms raw data into* ***clear, compelling insights****.*

Anscombe's Quartet



Four datasets with **same simple descriptive statistics**

- $\bar{x} = 9, s_x^2 = 11$
- $\bar{y} = 7.5, s_y^2 = 4.125$
- Correlation: $\rho_{xy} = 0.816$
- OLS: $y = 3 + 0.5x$

I		II		III		IV	
x	*y*	*x*	*y*	*x*	*y*	*x*	*y*
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

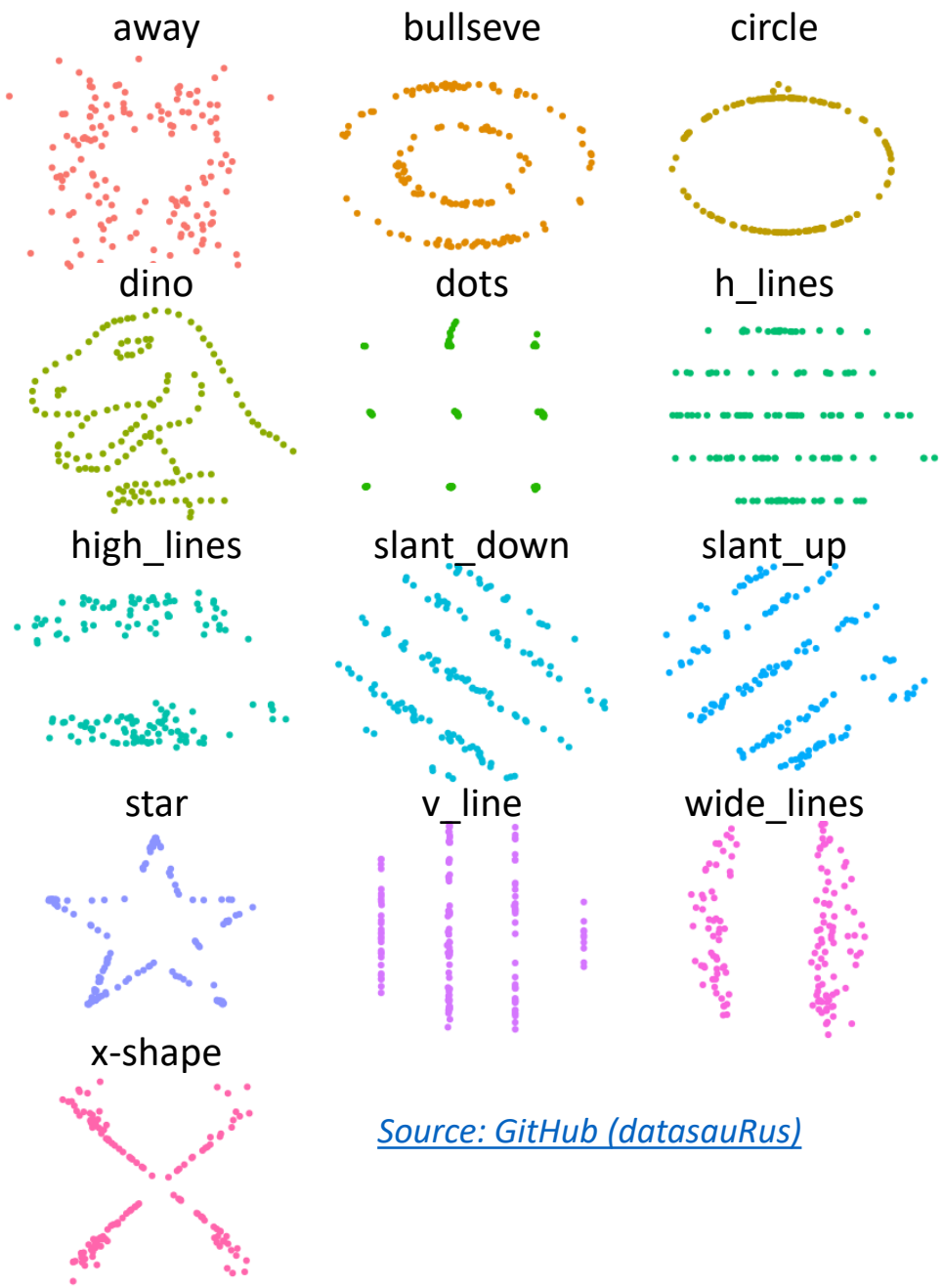
Source: Wikipedia (Anscombe's quartet)

Takeaway: *Numbers look the same → Visuals tell a different story.*

Datasaurus Dozen

- Inspired by Anscombe's quartet, 1973
- 13 datasets with nearly identical simple descriptive statistics (to two decimal places)
- Yet have very different distributions and appear very different when graphed

Source: Wikipedia (Datasaurus dozen)



Source: GitHub (datasauRus)

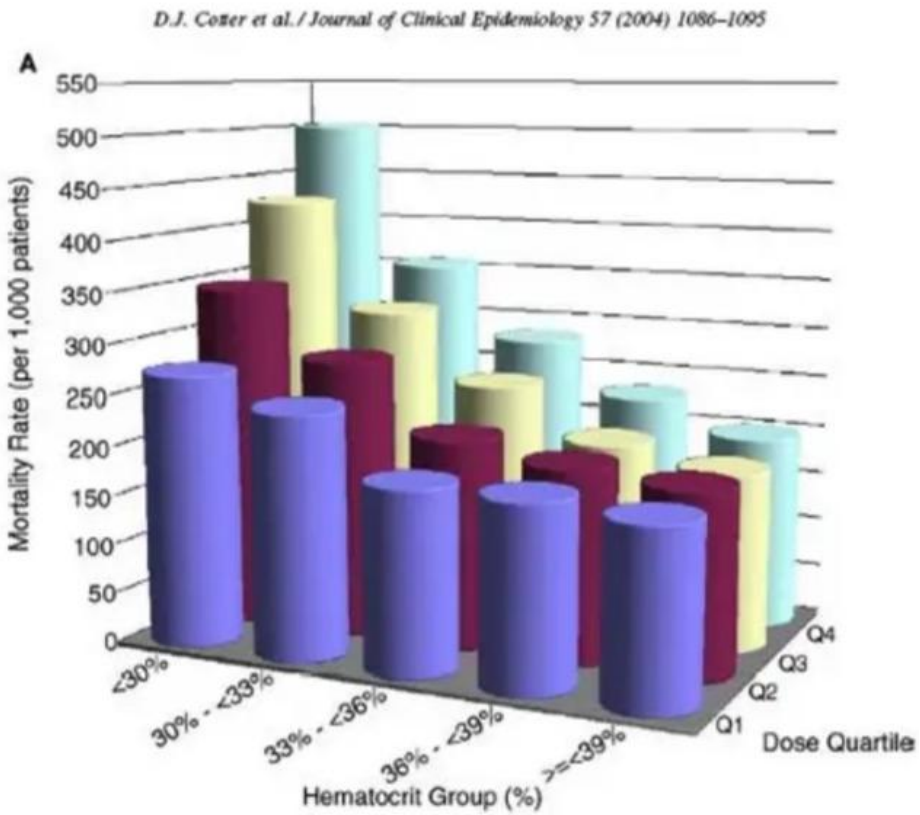
Takeaway: *Visualization is important – summary statistics can be the same but distributions can be very different.*

Lessons from Anscombe's Quartet & Datasaurus Dozen

- Statistics ≠ Distribution
  - Mean, variance, or correlation **cannot fully** represent the data
- Visualization Reveals Patterns
  - Shows relationships, outliers, and nonlinear structures **hidden in numbers**
- Outliers Matter
  - A few unusual points can **strongly** affect analysis and conclusions
- Key Insight
  - Relying only on statistics may mislead; visualization is **essential** for real understanding

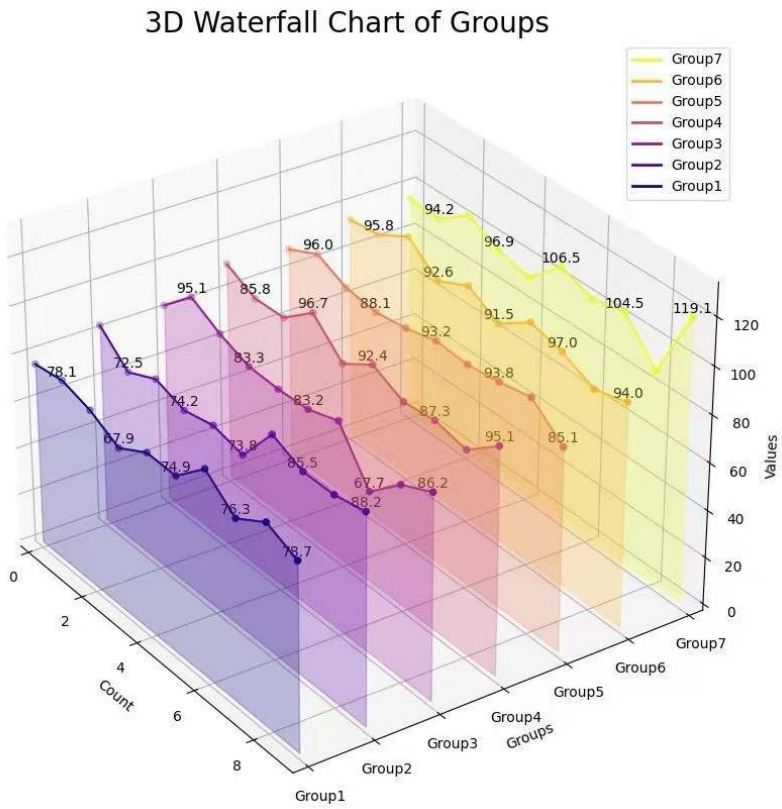
- **Descriptive Analytics**
  - Transforms raw data into information to understand past and current performance
  - Involves data **extraction**, **manipulation**, and **summarization** with statistical measures
  - Uses data visualizations to reveal **patterns** and **insights**

What Makes a Good Data Visualization?



Source: 5 examples of bad data visualization

3D? → hard to read



"beauty"? → 8/10?

practicability → **0/10**

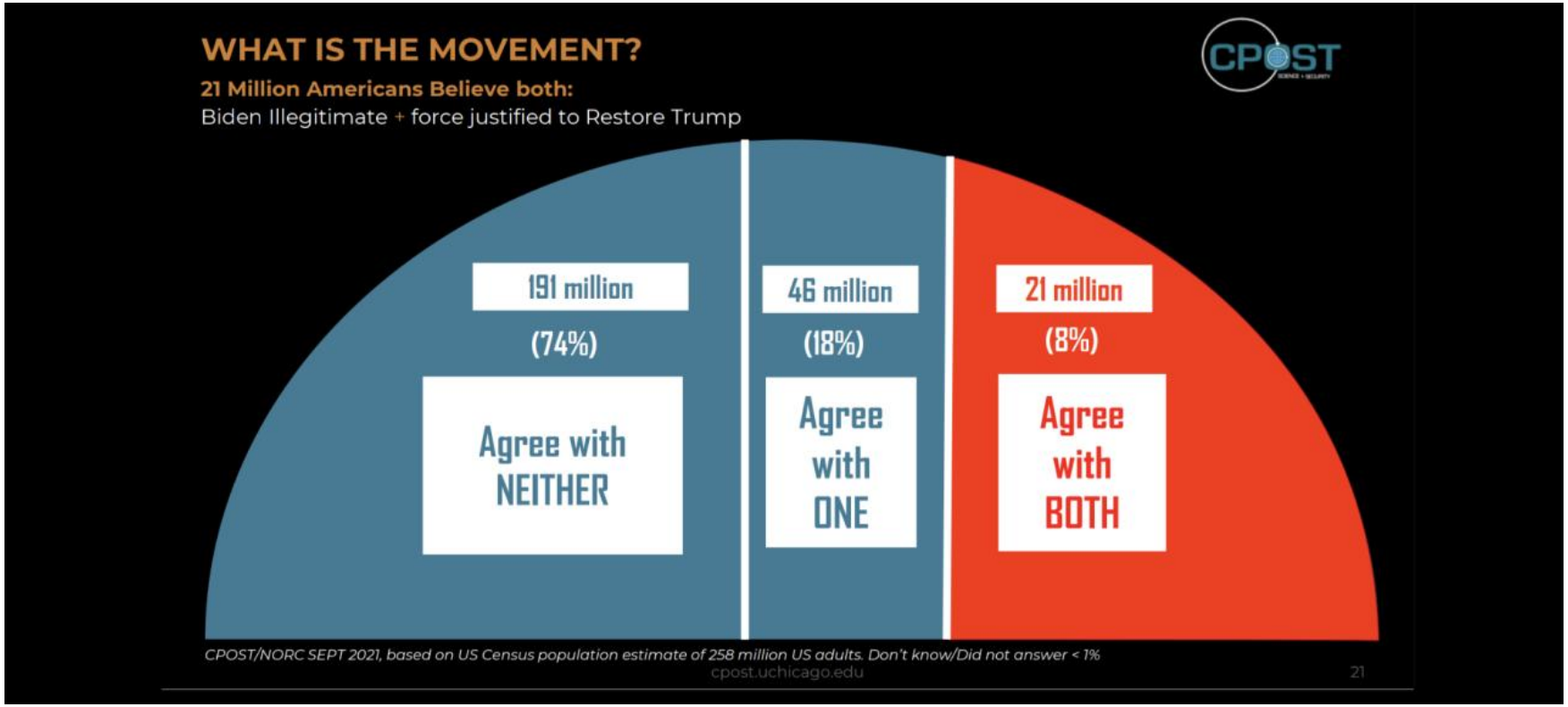
What Makes a Good Data Visualization?

Source: 5 examples of bad data visualization

line chart? discrete vs. continuous

wrong chart

What Makes a Good Data Visualization?



Source: WTF Visualizations

Any problem with this visualization?

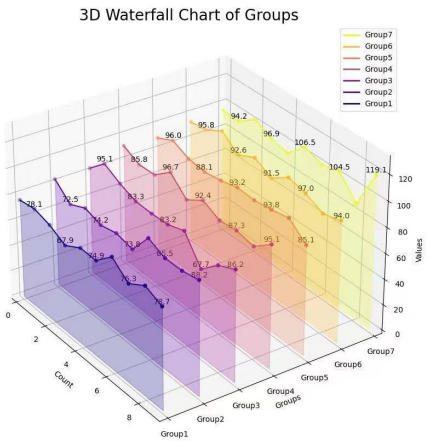
visual area ≠ actual percentages

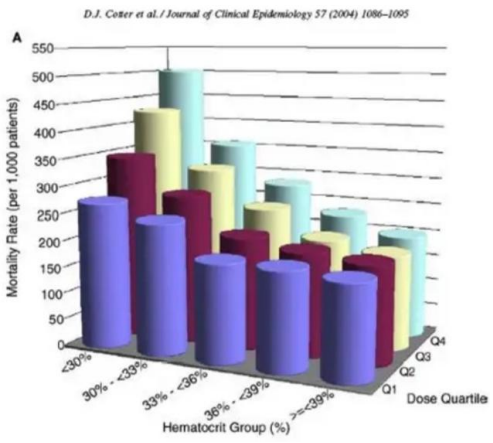
What Makes a Good Data Visualization?

- Know Your Audience
  - Design for their **needs** and **context**
- Be Accurate & Maintain Integrity
  - Represent data **truthfully**, avoid misleading scales
- Keep It Simple
  - Make it **easy to read**, minimize clutter and distractions
- Choose the Right Chart
  - **Match** visualization type to the data and question (trend, comparison, relationship, etc.)
- Use Design Wisely
  - Adopt **color**, **size**, and **shape** to emphasize patterns and reduce cognitive load
- Tell A Story
  - Highlight the key message and guide the audience to the **insight**

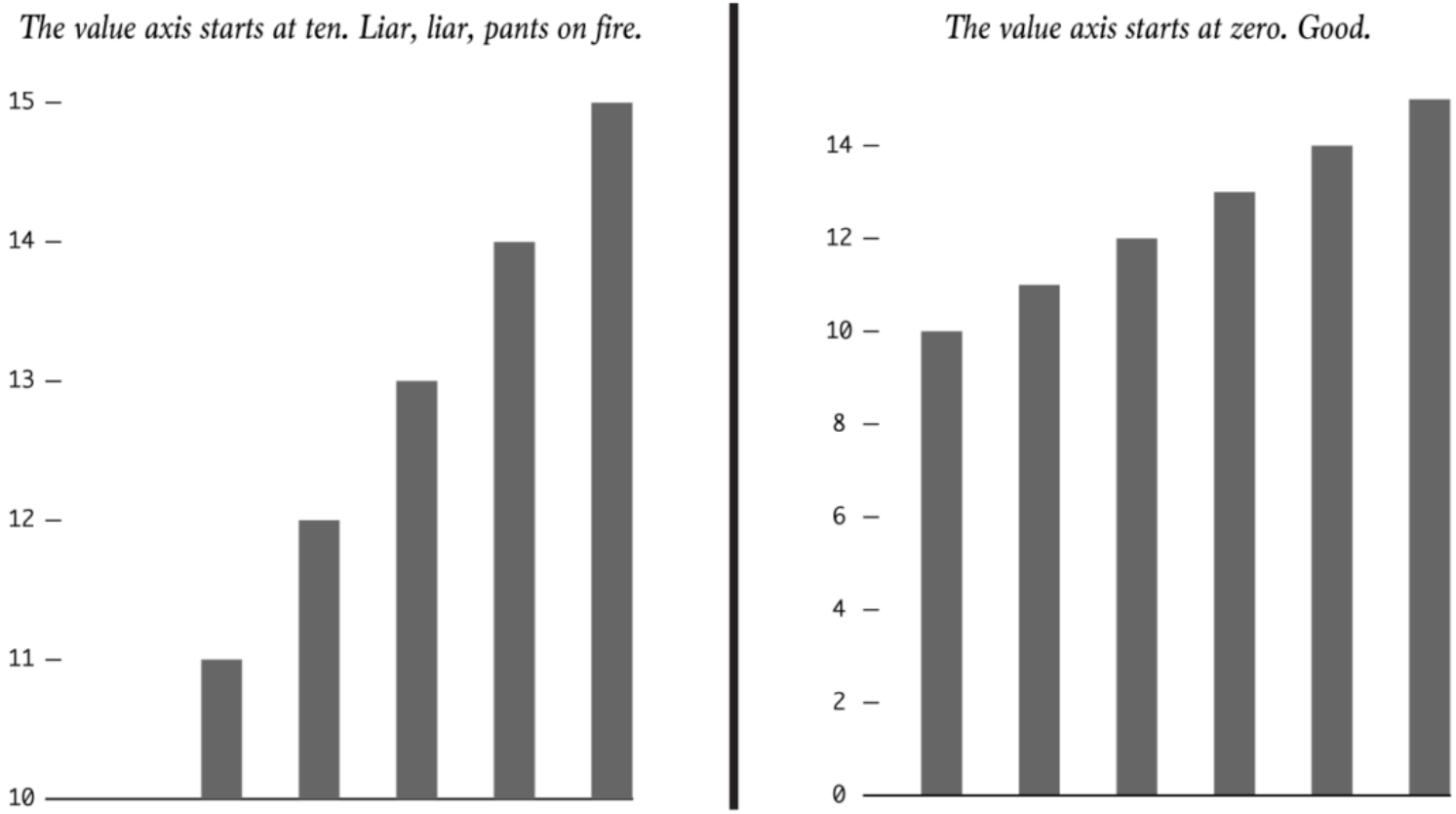
Common Pitfalls of Bad Visualization

- Misleading Scales
  - Truncated axes exaggerate differences
- Too Much Clutter
  - Unnecessary 3D effects, heavy gridlines, excessive labels
- Wrong Chart Type
  - E.g., pie chart for too many categories
- Overuse of Colors
  - Confusing palette or poor contrast
- Ignoring Context
  - Leaving out baseline, labels, or source
- Cherry-picking Data
  - Showing only a subset to push a narrative





Some Bad Visualizations

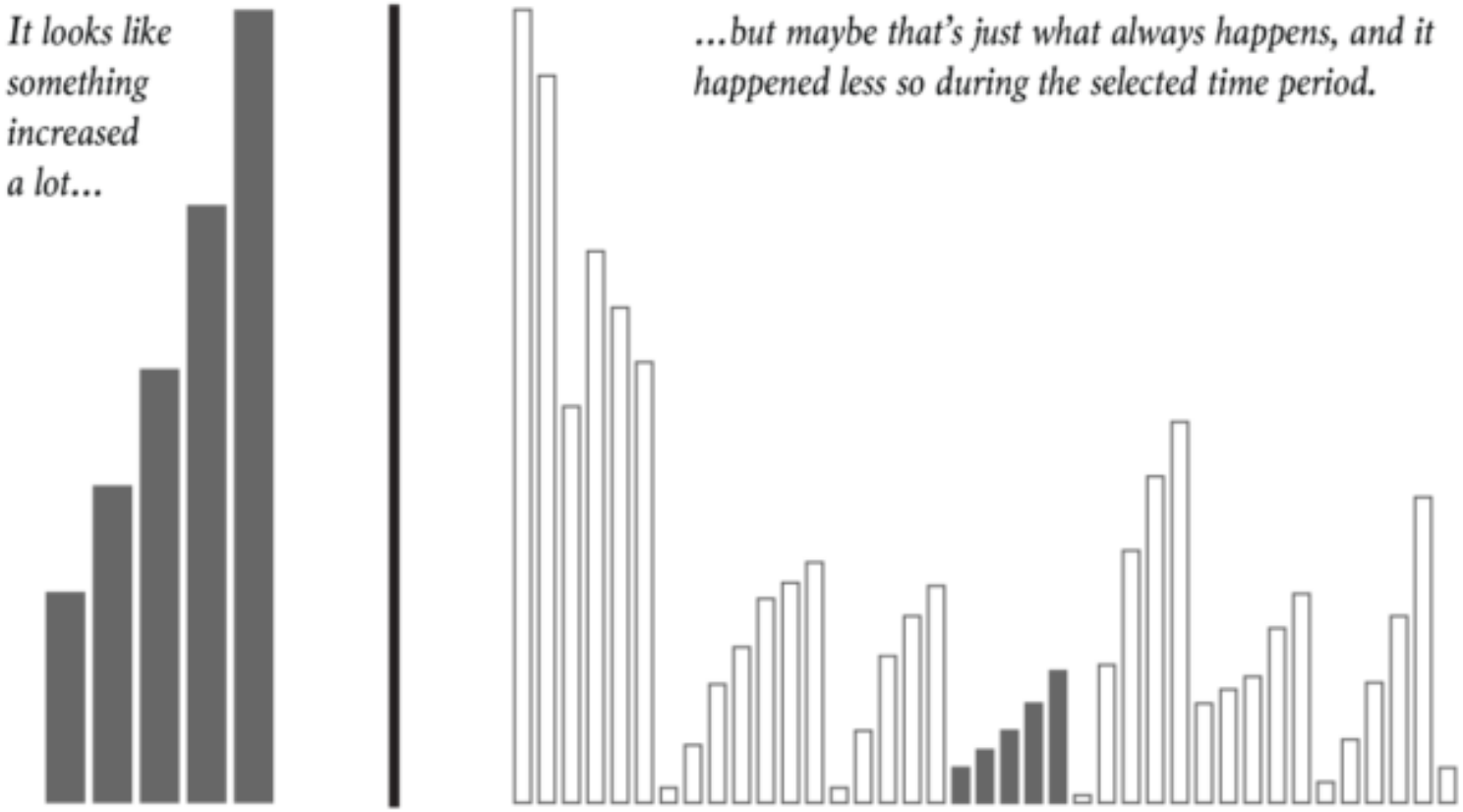


Source: How to Spot Visualization Lies

Any problem?

Bar charts use **length** as their visual cue, so when someone makes the length shorter using the same data by truncating the value axis, the chart dramatizes differences. Someone wants to show a bigger change than is actually there.

Some Bad Visualizations



Source: How to Spot Visualization Lies

Any problem?

It's easy to cherrypick **dates** and **timeframes** to **fit a specific narrative**. So consider history, what usually happens, and **proper baselines** to compare against.

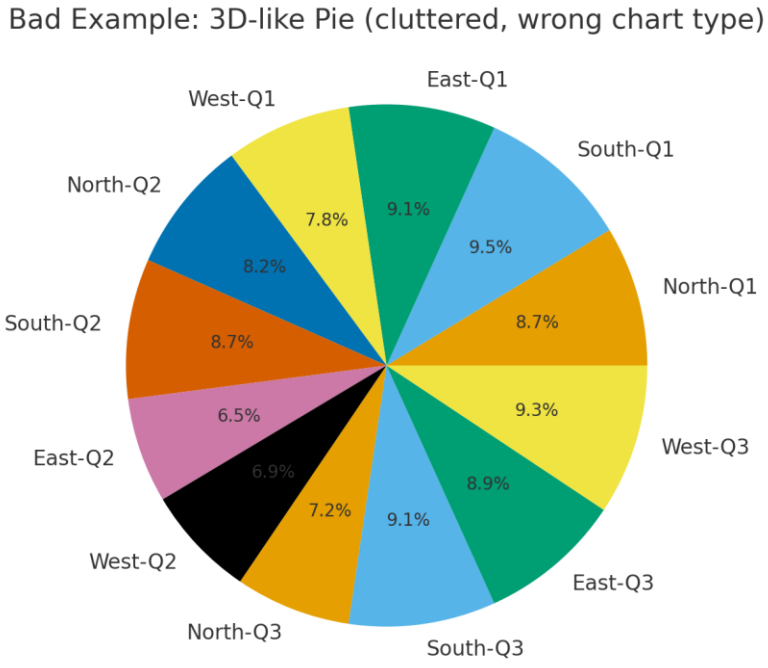
Think:

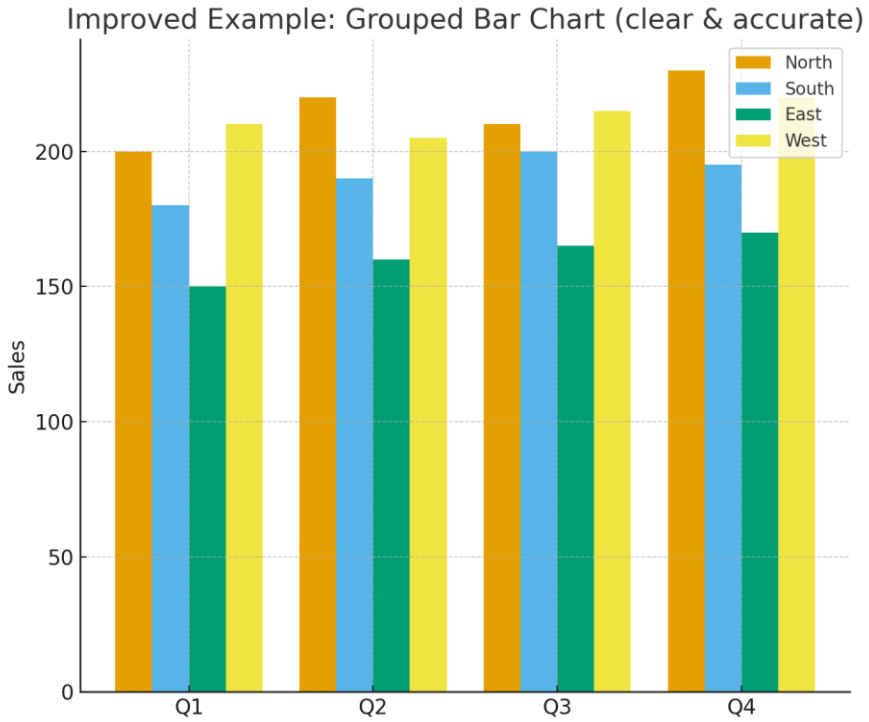
Intentional vs. **Unintentional**?

*The "**art**" of data visualization?*

Practice: A Mockup Chart

Scenario: A company wants to show quarterly sales for 4 regions.





Generated by ChatGPT5

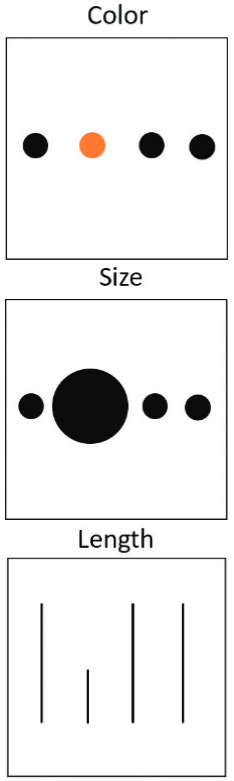
- Which pitfalls do you see in the pie chart?
- Why is the bar chart better?
- What other improvements could you suggest?

Using Design Wisely – Preattentive Attributes

- **Cognitive load** is the effort needed to accurately and efficiently process information communicated by a data visualization.
  - A data visualization with a reduced cognitive load is easier for the viewer to interpret.
- **Preattentive attributes** are features that can be used in a data visualization to reduce the cognitive load required to interpret it.
  - Preattentive attributes related to visual perception include:
    - Color
    - Size
    - Shape
    - Length

Using Design Wisely – Preattentive Attributes

- **Color** is extremely effective in differentiating data in a visualization
  - Yet overuse of colors may become distracting
  - Color blindness may affect the ability to distinguish colors
- **Size** refers to the relative amount of space an object occupies in a visualization
  - Most people struggle to estimate 2D relative size differences
- **Length** refers to the horizontal, vertical, or diagonal distance of a line or bar/column
  - Length is useful for illustrating quantitative values because a longer line corresponds to a larger value



Source: Camm Ch3

Using Design Wisely – Preattentive Attributes

The preattentive attributes of **color** and **size** aid with the counting of 7s

7	3	4	1	3	4	5	6	4	0
3	0	6	9	0	4	5	8	6	3
2	7	2	2	9	9	4	5	2	1
2	2	4	5	2	0	9	2	0	4
2	4	0	7	6	9	3	0	0	4
7	7	8	9	2	6	7	2	4	7
6	1	3	3	2	1	4	4	9	0
3	6	6	2	7	5	5	2	5	4
1	1	4	0	6	3	4	0	5	1
3	7	5	2	7	5	7	7	3	9
3	3	8	6	9	5	5	3	6	4
7	6	0	3	0	9	9	0	2	9
4	6	9	4	8	2	6	5	8	3
9	3	9	2	2	8	4	3	9	8
5	8	8	2	9	1	2	4	8	5
1	7	4	0	1	1	9	9	5	8

7	3	4	1	3	4	5	6	4	0
3	0	6	9	0	4	5	8	6	3
2	7	2	2	9	9	4	5	2	1
2	2	4	5	2	0	9	2	0	4
2	4	0	7	6	9	3	0	0	4
7	7	8	9	2	6	7	2	4	7
6	1	3	3	2	1	4	4	9	0
3	6	6	2	7	5	5	2	5	4
1	1	4	0	6	3	4	0	5	1
3	7	5	2	7	5	7	7	3	9
3	3	8	6	9	5	5	3	6	4
7	6	0	3	0	9	9	0	2	9
4	6	9	4	8	2	6	5	8	3
9	3	9	2	2	8	4	3	9	8
5	8	8	2	9	1	2	4	8	5
1	7	4	0	1	1	9	9	5	8

7	3	4	1	3	4	5	6	4	0
3	0	6	9	0	4	5	8	6	3
2	7	2	2	9	9	4	5	2	1
2	2	4	5	2	0	9	2	0	4
2	4	0	7	6	9	3	0	0	4
7	7	8	9	2	6	7	2	4	7
6	1	3	3	2	1	4	4	9	0
3	6	6	2	7	5	5	2	5	4
1	1	4	0	6	3	4	0	5	1
3	7	5	2	7	5	7	7	3	9
3	3	8	6	9	5	5	3	6	4
7	6	0	3	0	9	9	0	2	9
4	6	9	4	8	2	6	5	8	3
9	3	9	2	2	8	4	3	9	8
5	8	8	2	9	1	2	4	8	5
1	7	4	0	1	1	9	9	5	8

Source: Camm Ch3

Using Design Wisely – Data-Ink Ratio

- **Data-ink ratio** is the proportion of ink used for data to the total amount of ink in a table or chart
  - The data-ink ratio is helpful for creating effective tables and charts for data visualization
  - **Data-ink** is the ink used in a table or chart that is necessary to convey the meaning of the data to the audience
  - **Non-data-ink** is the ink used in a table or chart that serves no useful purpose in conveying the data to the audience
  - **Decluttering** is the process of increasing the data-ink ratio in a chart

Next, we consider two examples of decluttering in tables and charts for a data sample of scarf sales

Using Design Wisely – Data-Ink Ratio

low data-ink table

Scarf Sales			
Day	Sales (units)	Day	Sales (units)
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230

high data-ink table

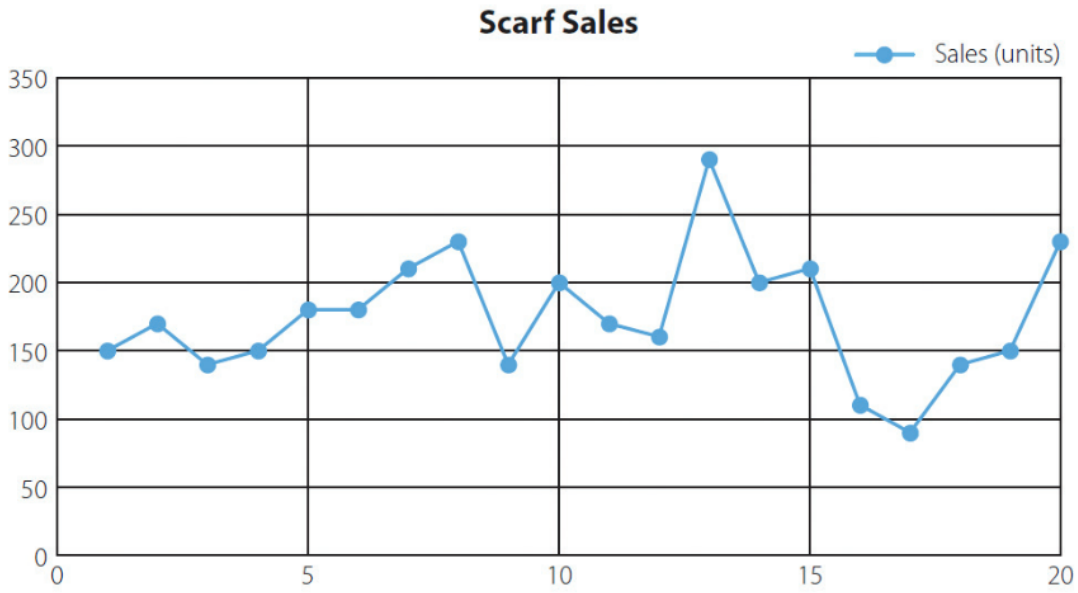
Scarf Sales			
Day	Sales (units)	Day	Sales (units)
1	150	11	170
2	170	12	160
3	140	13	290
4	150	14	200
5	180	15	210
6	180	16	110
7	210	17	90
8	230	18	140
9	140	19	150
10	200	20	230

Source: Camm Ch3

- **White space:** the portion of a data visualization devoid of markings
  - White space use is equivalent to increasing the data-ink in a visualization

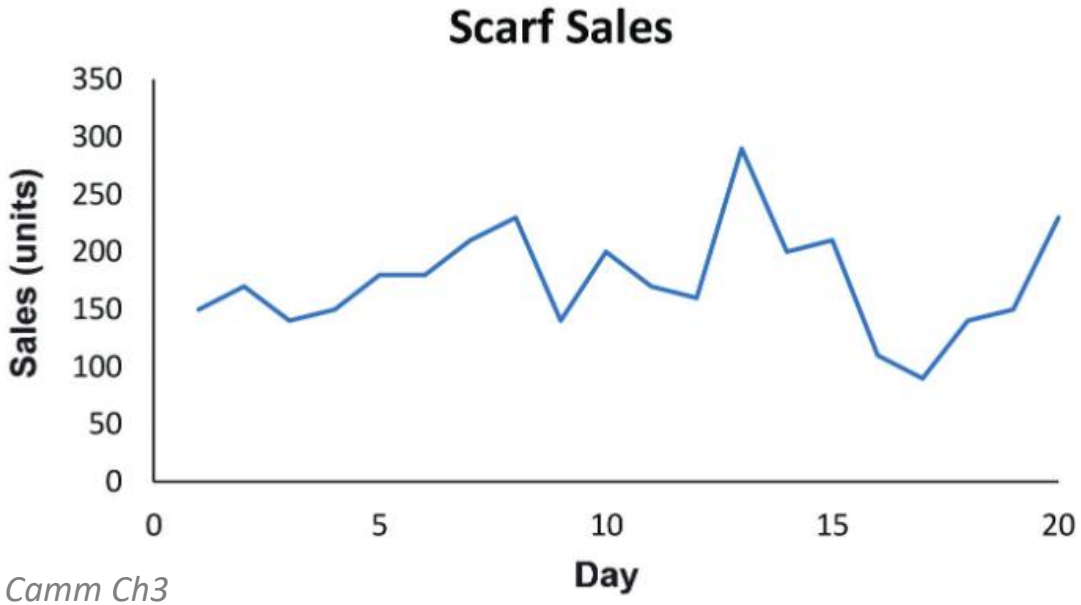
Using Design Wisely – Data-Ink Ratio

low data-ink chart



Source: Camm Ch3

high data-ink chart



- Always refine auto-generated visuals — remove clutter, keep it clear.

Data Visualization – Tables

- When to Use Tables
  - To provide specific numerical values
  - To allow precise comparisons between values
  - When values have different units or very different magnitudes

Example: Table of monthly costs and revenues for *Gossamer Industries*

	Month						
	Jan	**Feb**	**Mar**	**Apr**	**May**	**June**	**Total**
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,125	67,125	48,178	51,785	55,678	353,015

Source: Camm Ch3

Data Visualization – Tables

Table Design Tips

- Minimize clutter → keep the data-ink ratio high
- Use lines sparingly → only to separate labels, data, and calculated fields
- Align consistently
  - Labels: left-aligned
  - Values: right-aligned
  - Vertical labels: centered

Source: Camm Ch3

Design A:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design B:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design C:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Design D:

	Month						
	1	2	3	4	5	6	Total
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,128	67,125	48,178	51,785	55,687	353,027
Profits ($)	16,001	9,670	3,000	(3,980)	(2,933)	4,702	26,460

Data Visualization – Tables

Suppose that you wish to display data on revenues, costs, and head count for each month

- Costs and revenues are measured in **dollars**, but head count is measured in **number of employees**
- Because of the **different levels of magnitudes**, a table is preferred over a chart

	Month						
	Jan	**Feb**	**Mar**	**Apr**	**May**	**June**	**Total**
Head count	8	9	10	9	9	9	
Costs ($)	48,123	56,458	64,125	52,158	54,718	50,985	326,567
Revenues ($)	64,124	66,125	67,125	48,178	51,785	55,678	353,015

Source: Camm Ch3

Data Visualization – Crosstabulation (Cross-tab / Contingency)

- A method to summarize the relationship between two (or more) **categorical** variables
- Presents data in a **matrix** (table) format
  - rows = categories of one variable
  - columns = categories of another
- Shows **frequency counts** (or **percentages**) for each category combination
- Helps reveal patterns, relationships, or associations between variables

	Meal Price				
Quality Rating	**$10-19**	**$20-29**	**$30-39**	**$40-49**	**Total**
Good	42	40	2	0	84
Very Good	34	64	46	6	150
Excellent	2	14	28	22	300
Total	78	118	76	28	300

Source: Camm Ch3

Data Visualization – Charts

Basic Principles of Chart Selection

- Define the purpose
  - Clarify the goal: comparison, distribution, relationship, trend, etc.
- Know your data
  - Data type: categorical, ordinal, numerical, time series
  - Data dimensions
- Consider your audience
  - Match the chart design to their background and familiarity with the data
- Keep it simple
  - Choose the clearest, most direct chart type to convey information
- Avoid misleading visuals
  - Prevent distortion (e.g., truncated axes, inappropriate proportions)

Data Visualization – Charts

Recommended Charts by Analysis Purpose

Analysis Purpose	Recommended Chart Types
Compare values across categories	Column chart, bar chart, radar chart
Show **part-to-whole** relationships	Pie chart, donut chart, stacked column chart, tree map
Analyze data **distribution**	Histogram, box plot, scatter plot, density plot, bubble plot, heat map
Display **trend**s over time	Line chart, area chart, candlestick chart
Show **relationships & correlations**	Scatter plot, bubble chart, hear map, correlation matrix
Display **geographic** data	Map, heat map, point map, flow map
Display **hierarchical structure**	Tree map, sunburst chart, Sankey diagram
Show **network & connections**	Network graph, chord diagram, force-directed graph

Beyond the above basics, many other chart types and variations exist.

Summary

- Why Visualize?
  - Simplifies complexity, reveals patterns, supports decisions
- Lessons Learned
  - Statistics alone can mislead; visuals expose hidden insights
- Good Visualization
  - Clear, accurate, purposeful, and ethical
- How To Visualize
  - Use tables for precision, charts for patterns and comparisons
- Design Matters
  - Apply preattentive attributes, reduce clutter, maintain integrity

- Next class: How to draw tables and figures in Python (pandas, matplotlib, seaborn).
- After NDH: Assignment 1 grading, assignment 2 instructions & DDL, midterm exam (computer test) arrangement

Happy National Day Holiday!

- Wishing you a joyful and relaxing 8-day holiday
- Take time to **rest**, **recharge**, and **celebrate**
- Stay safe, enjoy with family and friends